USER MANUAL

Calibration targets · general user manual

General handling, capture and setup guidance. Use this document with the applicable family manual and datasheet; serial-specific values belong to the inspection report.

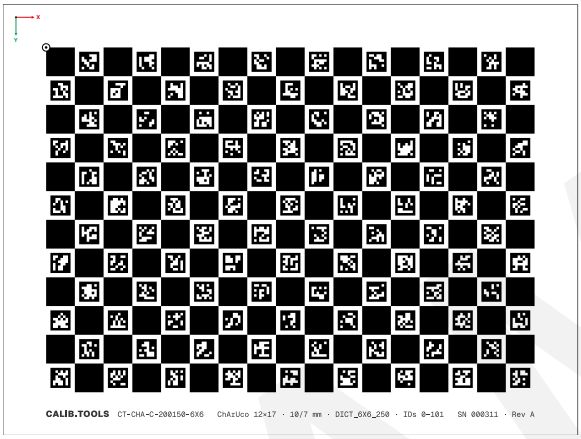
1 About this manual & safety

Use targets only as stable visual references. Handle the board with clean, dry hands on a clean padded surface; stop using it after a crack, chip in the pattern area or other visible change.

> SAFE FIRST USE
>
> **Before a critical run.** Confirm the part number, serial number, pattern definition and current camera configuration. This general manual does not replace the material- or mounting-specific family manual.

2 Identification

Read the board label before creating a target definition. The sample board below identifies every field.

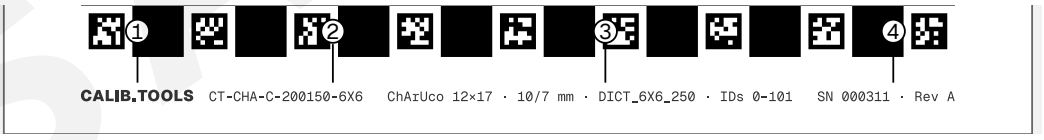


Rendering · sample board with printed X/Y axes and origin.

FIELD GUIDE

MARK	LABEL FIELD	READ / USE
1	Wordmark	CALIB.TOOLS identifies the maker.
2	Part number	CT-CHA-C-200150-6X6 identifies the target design.
3	Pattern summary	ChArUco 12 × 17 · 10 / 7 mm · DICT_6X6_250 · IDs 0–101.
4	SN · Rev	SN 000311 · Rev A identifies this board and its marking revision.

Origin (0, 0) is the top-left outer corner of the pattern; X points right and Y points down in the OpenCV frame.



Bottom-edge label detail · markers 1–4 correspond to the field guide above.

For the illustrated serial, open calib.tools/verify/000311 and enter 000311 to look up the associated inspection report and available records.

3 Unpacking & first check

CHECK	BEFORE FIRST USE	IF NOT ACCEPTABLE
Board and edges	Pattern face clean and dry; no chip, crack or deep scratch in the pattern area.	Do not mount or capture; photograph the condition.
Label and axes	Part number, SN, revision, X/Y arrows and origin are legible.	Record the serial number and contact support.
Documents	Match the inspection-report serial, when a report is supplied, to the board label.	Keep the delivery packaging and request guidance.

> IF SOMETHING IS WRONG
>
> Keep the original protective packaging until the first check is complete. For a concern, send the part number, serial number, photographs and the delivery reference to hello@calib.tools.

SAMPLE

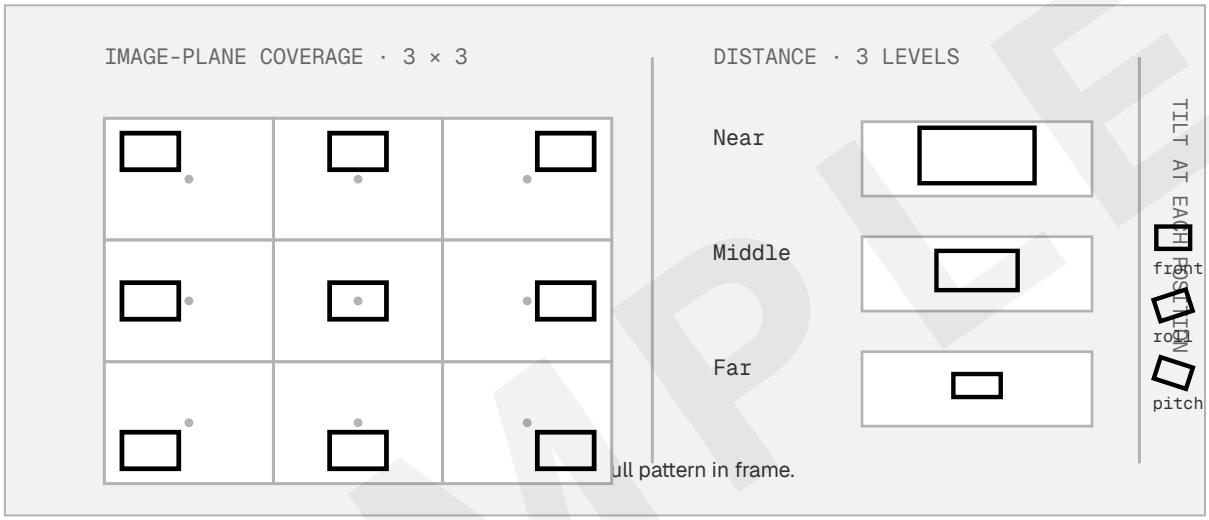
4 Handling & mounting

Support the board over a clean, stable surface. Keep fingers, tape, clamps and fixture features outside the active pattern area; consult the applicable family manual before using any dedicated mounting method.

STEP	DO	AVOID
1	Lift with two hands and keep the printed face clear of contact.	Sliding the face across a bench or fixture.
2	Support the full board or the specified mounting reference.	Point loading, twisting or clamping across the pattern.
3	Recheck label visibility after installation.	Obscuring the X/Y axes, origin or label line.

5 Capturing images

Capture a varied set, not repeated centre-on views. Keep the full pattern visible with a quiet border, then repeat the coverage whenever lens, focus, aperture or resolution changes.



Coverage checklist · 9 image positions × near / middle / far distance × front / roll / pitch tilt.

- ☐ No clipped highlights on white squares
- ☐ Black squares remain separated from noise
- ☐ Focus holds at the image corners
- ☐ Motion blur is absent at the target edge

For a guided capture review, use `calib.tools/tools/capture`.

6 Pattern files

FILE	USE WITH	USE
`CT-CHA-C-200150-6X6.opencv.json`	OpenCV	Board geometry, dictionary and start ID for Python or C++.
`CT-CHA-C-200150-6X6.kalibr.yaml`	Kalibr	Target definition for Kalibr workflows.
`CT-CHA-C-200150-6X6.ros2.yaml`	ROS 2	Board settings for compatible ROS 2 calibration workflows.
`CT-CHA-C-200150-6X6.halcon.descr`	HALCON	Target description for HALCON setup.
`CT-DWG-CHA-C-200150.dxf`	CAD / fixture	Outline and mounting reference; not a camera-calibration file.
`CT-CSV-000311.csv`	Analysis	Measured coordinates for the identified serial, when supplied.

LAYOUT COMPATIBILITY

OpenCV 4.6 and later. The ChArUco layout changed for even row counts. For an older even-row board, use `setLegacyPattern(True)` in the matching board definition; do not apply it to the current illustrated layout.

Multi-board start-ID setup continues on the next page.

SAMPLE

6 Pattern files

continued

For a multi-board set, set `start_id` independently for every board; the value selects the first marker ID in its row-major sequence.

SN	START_ID	MARKER IDS	DEFINITION
000311	0	0–101	First board
000312	102	102–203	Second board

`DICT_6X6_250` contains only 250 IDs; the illustrated 12 × 17 board uses 102 markers, so only two such boards fit without overlap.

7 First calibration (OpenCV quick start)

Use a board definition that matches its label: columns × rows, square and marker length in metres, dictionary and start ID. The example below is for the sample board with 10 / 7 mm geometry.

```
import cv2

aruco = cv2.aruco.getPredefinedDictionary(cv2.aruco.DICT_6X6_250)
board = cv2.aruco.CharucoBoard((17, 12), 0.010, 0.007, aruco)
    detector = cv2.aruco.CharucoDetector(board)
    charuco_corners, charuco_ids, marker_corners, marker_ids = detector.detectBoard(image)
```

Collect only frames with stable detection and broad coverage. Pass the accepted `charuco_corners` and `charuco_ids` from multiple images to your chosen OpenCV camera-calibration workflow.

> KEEP THE RECORD
> The board is inspected; it is not a camera-calibration result. Record the lens, image size, software version and target serial alongside your camera results.

8 Care & storage

DO	AVOID
Remove loose dust with a clean, dry, non-abrasive lint-free cloth.	Abrasion, scraping or rubbing the pattern with a soiled cloth.
Store flat in its protective packaging, clean and dry, under the datasheet storage conditions.	Loose storage under other parts, prolonged pressure or an unsupported edge.
Use the applicable family manual before any wet-cleaning method.	Unspecified solvents, coatings or adhesive labels on the pattern.

9 Re-inspection

These are suggested planning intervals, not a statement of board condition. Shorten the interval when handling, environment or image-quality risk is higher.

ENVIRONMENT	SUGGESTED INTERVAL	ALSO REQUEST AFTER
Controlled indoor reference use	24 months	Impact, uncertain storage, visible change or a critical-use concern.
Regular indoor engineering use	12 months	Frequent handling, relocation or an unexplained image-quality change.
Production / frequent handling	6 months	A drop, clamp event, contamination, heat exposure or edge damage.

To request S1e re-inspection, email `hello@calib.tools` with the part number, serial number, latest report reference, use environment and any incident or image-quality concern.

> RETURN PACKAGING
> For transport, place the clean board in its original protective sleeve and rigid packaging, immobilise it against movement, protect all edges, keep the serial visible in the shipment record and do not tape over the pattern.

SAMPLE

10 Inspection report & online verification

The inspection report is linked to a physical serial number. Match its serial and part number to the board label, then retain the PDF with your camera-calibration record.

Worked example	`CT-IR-000311 for SN 000311`
Online lookup	`calib.tools/verify/000311`
Measured coordinates	`CT-CSV-000311.csv, when supplied`

11 Warranty, returns & support

Warranty scope and return authorization are [Warranty terms — to be decided]. Ask support for current return instructions before sending a board.

Support	`hello@calib.tools`
Prepare	Part number · serial number · report reference · description of use · clear photographs
Return address	Calib.Tools · Returns · [Street and number] · [Postcode] Munich, Germany

12 Documents & downloads

Document availability depends on the ordered target and option. Use the file page below for the current general-manual package, then use the serial lookup for serial-specific records.

DOCUMENT	USE	REFERENCE
`Datasheet`	Product geometry, material and environment limits.	`CT-DS-CHA-C · Rev A`
`General manual`	This handling and setup guide.	`CT-UM-GEN · Rev A`
`Family manual`	Material care and mounting details for the target family.	`CT-UM-CHA-C`
`Inspection report`	Serial-specific inspection record.	`CT-IR-000311`
`Measured coordinates`	Serial-specific measured-corner data, when supplied.	`CT-CSV-000311.csv`
`Application note`	Further application guidance, where supplied.	`CT-AN-004`

General-manual files: `calib.tools/files/CT-UM-GEN`

13 Troubleshooting

PROBLEM	CHECK IN ORDER
No corners detected	1. Select `DICT_6X6_250` and the matching rows / columns. 2. Check focus, exposure and glare before changing detector thresholds. 3. For a historical even-row board only, apply the legacy-pattern setting described in section 6.
High reprojection error	1. Complete the coverage diagram: edges, distances and tilt. 2. Remove blurred or clipped images; confirm square and marker units. 3. Keep one camera configuration and one target definition for the run.
Multi-board ID conflict	1. Read the IDs and `start_id` on every board label. 2. Use start IDs `0` and `102` for the illustrated set. 3. Do not allocate beyond the 250 IDs in `DICT_6X6_250`.
Glare on the board	1. Use broad, diffuse illumination and move the light, not the board identity. 2. Tilt the light path away from the camera while keeping the target pose. 3. Clean fingerprints with the applicable family-care method; do not apply coatings to the pattern.

NEED THE CURRENT MANUAL PACKAGE?

Scan the QR code or use `calib.tools/files/CT-UM-GEN`. Keep this manual with the target definition used for your camera-calibration project.

`calib.tools/files/CT-UM-GEN`

SAMPLE